

Forecasting Volatility and Tail Risk During Extreme Market Events

Working Literature Review and Source Map

GARCH-EVT | Realized GARCH | Quantile Regression

Prepared for internal team review and paper verification

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Status of this document

This is a working scholarly synthesis built around the actual empirical project and its current code/results architecture. It is intended for team verification, not as the final submitted literature review. Bibliographic links are supplied so that teammates can verify the original papers. Where a paper has not yet been checked line-by-line from the full PDF, exact sample details and strong empirical claims should be confirmed before final submission.

Executive summary

The central intellectual question. The project is best understood not as a contest among three isolated econometric models, but as a comparison among three different ways of extracting information about the next-day conditional return distribution, especially its lower tail. GARCH-EVT models conditional scale and then extrapolates the residual tail; Realized GARCH enriches the volatility state using intraday realized measures; quantile regression estimates the conditional tail boundary directly without requiring a full return density.

The core conceptual distinction

A model can forecast conditional variance well and still forecast the 1% return quantile poorly. Conversely, a tail model can improve extreme-quantile calibration without changing the underlying volatility forecast. The literature therefore supports separating the paper into two evaluation lanes: volatility forecasting and tail-risk forecasting.

The literature changes the novelty claim. There is already substantial work combining GARCH with EVT, realized measures with GARCH, realized information with quantile models, and even hybrid systems that combine realized measures, EVT and conditional quantiles. The defensible contribution is therefore primarily empirical and comparative: a harmonized six-market out-of-sample experiment that separates volatility accuracy from tail calibration, studies performance during stress regimes, and treats realized-data quality as part of the forecasting problem.

The literature also implies several methodological checks. The strongest are: a skew-t Realized-GARCH robustness model; causal/strictly out-of-sample Realized-GARCH estimation; correct treatment of realized-measure scaling; EVT exceedance-dependence diagnostics; HAC-consistent Diebold-Mariano inference; and careful handling of Expected Shortfall because ordinary quantile regression does not itself produce ES.

Contents and literature architecture

Section	Purpose	Role in the argument
2.1	Stylized facts and conditional heteroskedasticity	Why volatility must be modelled dynamically.
2.2	GARCH, asymmetry and heavy-tailed conditional densities	Why GJR/EGARCH and skew-t are strong baselines.
2.3	Extreme Value Theory and GARCH-EVT	Why a second residual-tail stage may add information.
2.4	Realized volatility and high-frequency measurement	Why intraday data improve volatility measurement and what can go wrong.
2.5	Realized GARCH	How latent variance and observed realized measures are linked jointly.

2.6	Conditional quantiles and direct tail-risk models	Why the tail can be estimated directly.
2.7	Forecast evaluation: volatility, VaR and ES	Why model ranking depends on the target and the loss/test.
2.8	Crisis regimes, structural breaks and model instability	Why average performance can hide state dependence.
2.9	Comparative and hybrid literature; research gap	What has already been done and what remains defensible.

1. The intellectual spine of the literature review

The literature review should build a single cumulative argument rather than presenting four disconnected textbook summaries.

- Returns are conditionally heteroskedastic, asymmetric and heavy-tailed; therefore volatility needs a dynamic model.
- Even after volatility filtering, standardized innovations can retain economically important tail departures; therefore an EVT residual-tail layer may still matter.
- Daily squared returns are very noisy measures of latent variance; therefore high-frequency realized measures can improve information about the volatility state.
- The desired object in risk management is often a conditional quantile rather than the entire conditional density; therefore quantile regression supplies a fundamentally different modelling route.

The unifying empirical question is therefore: which information strategy produces the most reliable forecasts of conditional volatility and extreme downside risk, and does the answer change during market stress?

$$r_{t+1} = \mu_{t+1} + \sigma_{t+1} Z_{t+1}$$

The three main model families disagree about where the difficult information lies. GARCH-EVT focuses on the conditional scale and the residual tail; Realized GARCH focuses on more accurate measurement of the latent scale; quantile regression attempts to learn the tail boundary directly.

2.1 Stylized facts and conditional heteroskedasticity

Engle (1982): ARCH as the starting point

Engle introduced autoregressive conditional heteroskedasticity: the conditional variance of a time series may vary systematically with past shocks even when the unconditional variance is finite. In a simple ARCH(q) model,

$$h_t = \omega + \sum_{i=1}^q \alpha_i \varepsilon_{t-i}^2$$

This is the conceptual foundation for treating market volatility as a forecastable state rather than a constant nuisance parameter. In the present project, the relevance is direct: clustering in absolute/squared returns and ARCH-type diagnostics motivate the entire conditional-volatility branch.

Bollerslev (1986): persistence through GARCH

Bollerslev generalized ARCH by adding lagged conditional variance. For GARCH(1,1),

$$h_t = \omega + \alpha \varepsilon_{t-1}^2 + \beta h_{t-1}$$

This compact recursion generates persistent volatility without requiring a very large ARCH order. The quantity $\alpha + \beta$ is often used as a persistence diagnostic, but in the present study it should be interpreted cautiously: later structural-break literature shows that near-unit persistence can partly reflect nonstationarity or regime changes rather than a single stable long-memory law.

Poon and Granger (2003): modelling is not forecasting

Poon and Granger review the volatility-forecasting literature and make a distinction that should frame the report: good in-sample fit is not equivalent to good out-of-sample forecasting. Their review also emphasizes that the benchmark used as "actual volatility" matters enormously. This supports the project's emphasis on common out-of-sample targets and realized-volatility proxies rather than relying on likelihood rankings alone.

2.2 GARCH asymmetry and heavy-tailed conditional densities**GJR-GARCH and asymmetric news responses**

Equity volatility typically responds more strongly to negative than to positive return shocks of the same magnitude. Glosten, Jagannathan and Runkle (1993) formalize this asymmetry using the threshold term

$$h_t = \omega + \alpha \varepsilon_{t-1}^2 + \gamma I(\varepsilon_{t-1} < 0) \varepsilon_{t-1}^2 + \beta h_{t-1}$$

so that a negative innovation has marginal variance effect $\alpha + \gamma$ rather than α . This is the direct theoretical basis for the project's primary GJR filter.

Engle and Ng (1993): news-impact curves and sign-bias tests

Engle and Ng develop the news-impact curve and sign/size-bias diagnostics to test whether a volatility model captures asymmetric responses to news. Their empirical comparison finds strong asymmetry and particularly good performance from the GJR specification. This makes Engle-Ng a load-bearing citation for the project: asymmetry is not included merely because it is fashionable; it is motivated by explicit diagnostics.

Nelson (1991): EGARCH as an alternative asymmetry mechanism

EGARCH models log variance and allows signed standardized shocks to affect volatility directly. The log specification naturally avoids positivity constraints and provides an important alternative to threshold GJR dynamics. Because EGARCH-skew-t performs strongly in the project's in-sample information-criterion comparison, Nelson (1991) should be used to justify it as a serious robustness benchmark rather than a decorative extra model.

Hansen (1994): skewed Student-t innovations

A Gaussian innovation distribution is generally too light-tailed for equity returns. A Student-t allows excess kurtosis, but a symmetric t cannot reproduce left-right asymmetry. Hansen's skewed

Student-t conditional density allows both features. This matters greatly for the project because the production GJR-skew-t benchmark is intentionally strong: any incremental EVT benefit is being tested against a model that already captures leverage, fat tails and skewness.

Interpretive implication

If EVT adds only a modest improvement over GJR-skew-t, this should not be presented as a failed EVT experiment. It means that a strong parametric benchmark already absorbs much of the tail behaviour that weaker Gaussian or symmetric-t GARCH benchmarks leave for EVT to explain.

2.3 Extreme Value Theory and GARCH-EVT

Why the Generalized Pareto distribution appears

The Pickands-Balkema-de Haan theorem gives the mathematical foundation for Peaks Over Threshold (POT). For a sufficiently high threshold u , the conditional distribution of exceedances $X-u$ given $X>u$ is approximated by a generalized Pareto distribution (GPD).

$$G_{\xi, \beta}(y) = 1 - (1 + \xi y/\beta)^{-1/\xi}$$

Balkema and de Haan (1974) and Pickands (1975) provide the asymptotic theory. Hill (1975) is useful for tail-index diagnostics. These sources justify the mathematical object; they do not tell the researcher exactly where to place a finite-sample threshold, which is why diagnostics and sensitivity analysis remain necessary.

Davison and Smith (1990): practical POT modelling

Davison and Smith systematize threshold exceedance modelling, including GPD estimation, threshold diagnostics, covariates and dependence. This is particularly relevant because the project should not assume that standardized extreme residuals are independent merely because ordinary volatility clustering has been filtered out. Exceedance-indicator dependence is itself a diagnostic worth reporting, especially where extreme events cluster.

McNeil and Frey (2000): the governing conditional-EVT framework

McNeil and Frey combine a volatility filter with EVT to estimate conditional VaR and ES. The logic is two-stage:

1. Estimate conditional mean and scale from a heteroskedastic model: $r_t = \mu_t + \sigma_t z_t$.
2. Standardize returns, transform the loss tail if necessary, and fit a GPD only to sufficiently extreme residual exceedances.
3. Map the estimated standardized extreme quantile back into return units using the next-period conditional mean and scale.

$$\text{VaR}_{\alpha, t+1} = \hat{\mu}_{t+1} + \hat{\sigma}_{t+1} \hat{q}_{\alpha, \text{EVT}}$$

The project's causal GARCH-EVT implementation belongs directly in this tradition. Importantly, GARCH-EVT does not create a new volatility forecast when it shares the same GJR filter: it changes the tail mapping. Therefore the empirical report should never compare GJR and GARCH-EVT as if they were distinct variance models.

Kuester et al. (2006), Jalal and Rockinger (2008), and Trapin (2018)

Kuester, Mittnik and Paolella show that heavy-tailed GARCH-EVT can perform strongly in large VaR comparisons. Jalal and Rockinger then provide an important caution: the filtered-EVT approach depends on the quality of the first-stage filter, and misspecified filtering can distort tail inference. Trapin shifts attention to extremal dependence itself, showing that extreme negative returns can display persistent dependence that ordinary volatility models need not reproduce. Together these studies justify residual diagnostics, squared-residual diagnostics, tail-threshold checks and explicit exceedance-clustering tests.

2.4 Realized volatility and high-frequency measurement

Andersen and Bollerslev (1998): the volatility proxy problem

A daily squared return is an extremely noisy proxy for the day's latent variance. Andersen and Bollerslev show that standard volatility models can look much more accurate when evaluated against high-frequency-based ex-post volatility measures. This is one of the central reasons the project evaluates variance forecasts against realized measures rather than treating r^2 as literal realized volatility.

Andersen, Bollerslev, Diebold and Labys (2003): realized volatility and quadratic variation

The realized-volatility literature connects sums of squared intraday returns to the quadratic variation of the underlying price process. For intraday returns $r_{t,1}, \dots, r_{t,M}$,

$$RV_t = \sum_{j=1}^M r_{t,j}^2$$

under appropriate sampling conditions provides a much more informative ex-post measure of daily variation than a single close-to-close squared return. This paper is therefore the principal foundation for the project's intraday realized-variance construction.

Microstructure noise: why the finest frequency is not automatically best

Observed high-frequency prices contain bid-ask bounce, discreteness, stale quotes and other microstructure effects. Zhang, Mykland and Ait-Sahalia (2005) show formally that realized variance at ever-finer sampling frequencies can be inconsistent in the presence of noise and develop a two-scale solution. Barndorff-Nielsen, Hansen, Lunde and Shephard (2008) develop realized kernels as another noise-robust approach.

This literature supports the project's use of a moderate 5-minute baseline together with frequency-sensitivity checks. It does not justify the claim that 5 minutes is universally optimal; rather, 5 minutes is a defensible practical compromise, and Liu, Patton and Sheppard (2015) provide strong cross-asset evidence that simple 5-minute RV is difficult to beat consistently.

Jumps, semivariances and robust realized measures

Barndorff-Nielsen and Shephard (2004) develop bipower variation, which separates continuous integrated variance from jump variation under suitable conditions. Barndorff-Nielsen, Kinnebrock and Shephard (2010) decompose realized variance into positive and negative semivariances. Andersen, Dobrev and Schaumburg (2012) develop MinRV/MedRV as jump-robust estimators. These

are not ancillary technicalities: they provide theory for the project's BPV, continuous/jump variation, jump share, RS-positive/negative, RSV ratio and MedRV features.

Corsi (2009): multiple volatility horizons

HAR-RV represents realized volatility through daily, weekly and monthly components. It reproduces long-memory-like dynamics through heterogeneous horizons without requiring fractional integration. This is the theoretical basis for the project's daily/weekly/monthly log-RV predictor block and helps explain why those variables should be pre-specified economically rather than discovered by unrestricted feature search.

2.5 Realized GARCH

Hansen, Huang and Shek (2012): joint return-measurement modelling

Realized GARCH differs from simply inserting lagged RV into ordinary GARCH. It models latent return variance and the observed realized measure jointly. A representative log-linear form is

$$\begin{aligned} r_t &= \mu + \sqrt{h_t} z_t \\ \log h_t &= \omega + \beta \log h_{t-1} + \gamma \log x_{t-1} \\ \log x_t &= \xi + \varphi \log h_t + \tau_1 z_t + \tau_2 (z_t^2 - 1) + u_t \end{aligned}$$

The measurement equation is the defining feature: x_t is an observed high-frequency volatility measure linked statistically to the latent h_t . The leverage function $\tau_1 z_t + \tau_2 (z_t^2 - 1)$ allows return shocks to affect the measurement asymmetrically. This paper is the governing reference for the project's Realized-GARCH implementation.

Shephard and Sheppard (2010): realized information and missingness

The HEAVY literature provides useful context even though HEAVY is not a fitted project model. It emphasizes rapid updating from high-frequency measures and explicitly discusses cases where realized observations are missing. This is particularly relevant to the Nikkei data-quality episode: realized-information models gain information when the measurement stream is good, but they also inherit an additional failure mode when the measurement stream degrades.

Watanabe (2012): the skew-t Realized-GARCH robustness check

Watanabe studies one-day VaR and ES from Realized GARCH under Gaussian, Student-t and skewed Student-t innovations. In the S&P 500 application, the skew-t Realized-GARCH version performs better than the symmetric alternatives and an EGARCH benchmark. This is now a mandatory paper for the project because the current Realized-GARCH return equation uses a symmetric Student-t even though fitted standardized residuals retain negative skew. A RealGARCH-skew-t robustness run is therefore directly motivated by prior literature rather than being an ad hoc response to the results.

Giot and Laurent (2004): better realized volatility need not mean better VaR

Giot and Laurent compare realized-volatility-based VaR with a strong skewed-Student APARCH benchmark. Their results are deliberately nuanced: realized information can generate adequate

risk forecasts but does not automatically dominate a well-specified daily asymmetric heavy-tailed model. This is exceptionally relevant to the project's likely empirical story that Realized GARCH can improve QLIKE substantially while still delivering poor 1% VaR calibration under an inadequate return-density specification.

Potential central empirical message

Accurate conditional-variance estimation is not sufficient for accurate extreme return quantiles. If the Realized-GARCH variance advantage survives strict out-of-sample refitting while its 1% VaR remains weak even after a skew-t innovation robustness check, this separation becomes one of the strongest scientific results in the paper.

2.6 Quantile regression and direct tail-risk forecasting

Koenker and Bassett (1978): estimate the quantile directly

Quantile regression changes the estimand from a conditional mean to a conditional quantile. For quantile level τ ,

$$\hat{\beta}_\tau = \arg \min_{\beta} \sum_t \rho_\tau(y_t - x_t' \beta)$$

with the asymmetric check loss ρ_τ . This makes quantile regression a genuinely different modelling philosophy. GARCH-based methods specify a conditional scale and distribution and then extract a quantile; QR estimates the desired quantile directly.

Engle and Manganelli (2004): CAViaR and the Dynamic Quantile test

CAViaR introduces dynamic conditional VaR models estimated by regression quantiles and proposes the Dynamic Quantile (DQ) test. A correctly specified α -level VaR should satisfy

$$P(r_t < \text{VaR}_{\alpha,t} \mid \mathcal{F}_{t-1}) = \alpha$$

and the hit sequence should not be predictable from information available to the forecaster. This provides the theoretical basis for using the DQ test in addition to unconditional and conditional coverage tests.

Žikeš and Baruník (2016): the closest foundation for QR-Full features

Žikeš and Baruník study conditional quantiles using realized variance, upside/downside semivariance, jump variation and option-implied volatility. Their framework is especially close to the project's QR-Full architecture and supplies an academic rationale for variables such as LogRV, RSV ratio, JumpShare and LogIV. The implication is important: these variables should be presented as economically and statistically motivated predictor blocks, not as an unrestricted data-mining exercise.

Implied volatility, variance risk premium and range predictors

Blair, Poon and Taylor (2001) show that option-implied volatility contains substantial forward-looking information. Bollerslev, Tauchen and Zhou (2009) give economic meaning to the variance risk premium: the wedge between option-implied and realized variation contains information about market risk. Parkinson (1980) and Garman and Klass (1980) provide classic OHLC-based

variance estimators. Together these sources support the project's implied-volatility/VRP variables and its range-only fallback QR specification.

Bee, Dupuis and Trapin (2018): hybrid realized + quantile + EVT models already exist

The Realized Extreme Quantile model combines realized measurement, conditional quantiles and EVT refinements. This is a critical novelty constraint: the paper must not claim that combining realized information, quantile modelling and EVT is unprecedented. Rather, the project's contribution is the controlled comparison of different information strategies within one harmonized multi-market experiment.

2.7 Forecast evaluation: volatility, VaR and Expected Shortfall

Volatility losses: Hansen-Lunde and Patton

Hansen and Lunde (2006) show that noisy volatility proxies can mis-rank competing models, including the possibility that squared daily returns select an inferior model with increasing probability. Andersen, Bollerslev and Meddahi (2005) emphasize measurement error in realized volatility itself. Patton (2011) then characterizes loss functions whose forecast rankings are robust under conditionally unbiased proxy noise. This literature strongly supports the project's use of realized variance together with QLIKE as a primary volatility criterion.

$$\text{QLIKE}(v,h) = v/h - \log(v/h) - 1$$

VaR coverage and dynamic adequacy

Kupiec (1995) tests whether the unconditional violation rate matches the nominal VaR level. Christoffersen (1998) adds independence and conditional-coverage requirements because a model can have the correct average number of breaches while still producing highly clustered failures. Engle and Manganelli's DQ test goes further by asking whether the hit process is predictable from a broader information set. These tests answer different questions and should therefore be reported separately.

Diebold-Mariano and Model Confidence Set

Diebold and Mariano (1995) compare predictive accuracy through the mean loss differential and explicitly allow serial correlation. Therefore the long-run variance of the loss differential should be estimated appropriately; a HAC/Newey-West treatment is the natural correction when losses are autocorrelated. Hansen, Lunde and Nason (2011) provide the Model Confidence Set, which is especially useful when many models are compared across markets and losses because it avoids forcing a single winner when several models are statistically indistinguishable.

Expected Shortfall: Fissler-Ziegel and modern joint VaR-ES models

Expected Shortfall cannot simply be treated as another quantile. Fissler and Ziegel (2016) show that the pair (VaR, ES) is jointly elicitable and admits strictly consistent joint scoring functions. Taylor (2019) and Patton, Ziegel and Chen (2019) use this insight to construct dynamic semiparametric VaR-ES models. The practical consequence for the project is straightforward: vanilla single-quantile

QR produces VaR but not ES. Either ES comparisons should be restricted to models that genuinely produce ES, or the QR branch must be extended to a joint VaR-ES model.

2.8 Crisis regimes, structural breaks and model instability

Lamoureux and Lastrapes (1990): apparent persistence can reflect breaks

Structural changes in unconditional variance can make GARCH persistence appear excessively close to one. This is an important qualification for the high persistence estimated in the project. High $\alpha + \beta$ is genuine evidence of persistent volatility dynamics, but it should not be interpreted as proof that one stationary volatility law governs the entire sample.

Mikosch-Stărică and Granger-Hyung: long memory versus nonstationarity

Mikosch and Stărică (2004) show that nonstationarity can generate apparent long-range dependence and IGARCH-like effects. Granger and Hyung (2004) show that occasional structural breaks can mimic slowly decaying autocorrelations. Therefore the project's GPH estimates should be described as long-memory-like persistence rather than proof of a unique fractional-memory data-generating process.

Pesaran and Timmermann (2007): why estimation-window length is an empirical choice

When breaks occur, discarding old data reduces bias but increases parameter-estimation variance. Pesaran and Timmermann formalize this trade-off. This gives a rigorous basis for the project's expanding versus fixed 2-year/5-year window sensitivity analysis. The right conclusion is not that expanding windows are automatically correct, but that window length is a bias-variance decision whose robustness should be demonstrated empirically.

These papers also justify keeping named crisis windows and ex-ante volatility regimes conceptually separate. A calendar event such as COVID identifies an economic episode; a volatility regime identifies the contemporaneous state of the conditional variance process. Both can be informative, but they answer different questions.

2.9 Comparative and hybrid literature: what has already been done

Louzis et al. (2014): volatility model quality and quantile method are separate choices

Louzis, Xanthopoulos-Sisinis and Refenes compare realized-volatility models and GJR-GARCH under several tail-quantile estimators, including skew-t and EVT. Their results emphasize that the quantile method can be as important as the volatility model. This maps almost perfectly onto the project's distinction between variance quality and tail-distribution quality.

Paul and Sharma (2017, 2018): Realized-GARCH-EVT already exists

Paul and Sharma explicitly combine Realized GARCH with EVT and compare it with GARCH-EVT/EGARCH-EVT alternatives for VaR and later VaR/ES. This rules out novelty claims based on simply combining realized information and EVT. In the present paper, Realized-GARCH-EVT is better positioned as a natural future extension or additional robustness model if scope permits.

Gerlach and Wang (2020): a very close broad comparator

Gerlach and Wang develop semiparametric joint VaR-ES models driven by realized measures and compare them with GARCH, Realized GARCH and EVT benchmarks over seven financial indices. This is one of the closest existing papers to the broad conceptual comparison and therefore places a hard constraint on any claim that the three modelling families have never been compared.

Recent realized tail-risk literature

Wu and Cai (2024) compare a variety of realized measures, including RV, MedRV, BPV and realized kernels, in joint VaR-ES models and use modern scoring/MCS tools. **Chen, Koike and Shau (2024)** show that overnight information can add to realized measures in international tail-risk forecasting. These papers are especially relevant to the project's international design, robust-realized-measure checks and the information-set timing problem for Asian markets.

3. Defensible research gap and contribution

The literature does not support a novelty claim of the form "GARCH-EVT, Realized GARCH and quantile regression have never been compared or combined." Variants and hybrids already exist. The defensible gap is more specific and, in many ways, more interesting.

Recommended research-gap wording

Existing research has established the merits of asymmetric heavy-tailed GARCH models, conditional EVT, realized-measure volatility models and semiparametric conditional-quantile approaches. A growing literature also develops hybrids that combine realized measures with EVT or direct VaR/ES dynamics. Less attention has been given to a common multi-market out-of-sample design that treats these as competing information philosophies - conditional scale plus residual-tail extrapolation, high-frequency measurement of latent volatility, and direct conditional-quantile prediction - while separately evaluating variance forecasting and tail-risk calibration, conditioning performance on market stress, and explicitly accounting for realized-data quality.

The contribution is therefore best described as empirical and comparative. The six-market US/Europe/Asia design, common forecast contract, explicit separation of volatility and tail objectives, crisis/regime analysis, causal residual-tail construction and data-quality robustness collectively provide the paper's identity.

4. Research questions implied by the literature

RQ1 - Volatility information: Does incorporating intraday realized measures improve one-step-ahead conditional-variance forecasts relative to a returns-only asymmetric GARCH benchmark?

RQ2 - Incremental EVT value: Does EVT materially improve extreme downside-risk forecasts after returns have already been filtered by an asymmetric skew-t GARCH model?

RQ3 - Direct quantiles: Can quantile regression using realized, implied, asymmetry and range information forecast lower conditional return quantiles as accurately as models derived from explicit return distributions?

RQ4 - State dependence: Do relative model rankings change during named crisis periods or ex-ante high-volatility regimes?

RQ5 - Information quality: Are gains from realized-measure models robust to degradation or absence of intraday information, especially in markets with session/feed discontinuities?

5. What the literature implies for the empirical project

Literature implication	Project consequence	Priority
Watanabe (2012): skew-t Realized GARCH can outperform symmetric-t versions for VaR/ES.	Run RealGARCH-skew-t robustness; otherwise poor VaR may be confounded by density misspecification.	High
McNeil-Frey + Jalal-Rockinger + Trapin: filtered extremes still require diagnostic adequacy.	Retain residual/squared-residual checks and add EVT exceedance-independence/declustering diagnostics.	High
Hansen-Lunde + Patton: volatility rankings depend on proxy/loss quality.	Keep realized variance and QLIKE as the primary variance-evaluation framework.	High
Diebold-Mariano allows serially correlated loss differentials.	Use HAC/Newey-West long-run variance in final DM inference.	High
Fissler-Ziegel: VaR and ES require joint scoring; vanilla QR does not produce ES.	Do not manufacture QR ES; either restrict ES comparison or add a joint VaR-ES model.	High
Microstructure-noise literature: finest sampling is not automatically best.	Keep 5-minute RV as baseline but retain frequency sensitivity; fix any subsampling implementation before citing it.	Medium
Structural-break literature: persistence/long memory may partly reflect nonstationarity.	Use cautious wording and retain expanding-vs-fixed window robustness.	Medium
HEAVY/realized-model literature: information quality matters.	Make the NKY missing/invalid realized-measure episode a substantive robustness/limitation, not a footnote.	High
Hybrid literature already combines realized measures, QR and EVT.	Keep novelty claim empirical/comparative rather than methodological invention.	High

6. Recommended literature corpus

The evidence matrix prepared alongside this review contains a broader verification corpus. For the written report, the following papers are the minimum serious scholarly backbone; other papers should be used selectively for implementation details, robustness and recent context.

Conditional volatility and asymmetry: Engle (1982); Bollerslev (1986); Nelson (1991); Glosten, Jagannathan & Runkle (1993); Engle & Ng (1993); Hansen (1994); Poon & Granger (2003).

EVT and conditional tail risk: Balkema & de Haan (1974); Pickands (1975); Hill (1975); Davison & Smith (1990); McNeil & Frey (2000); Kuuster, Mittnik & Paolella (2006); Jalal & Rockinger (2008); Trapin (2018).

Realized volatility and Realized GARCH: Andersen & Bollerslev (1998); Andersen et al. (2003); Zhang, Mykland & Ait-Sahalia (2005); Hansen & Lunde (2006); Corsi (2009); Hansen, Huang & Shek (2012); Liu, Patton & Sheppard (2015); Watanabe (2012); Giot & Laurent (2004).

Quantiles and tail-risk regression: Koenker & Bassett (1978); Engle & Manganelli (2004); Žikeš & Baruník (2016); Bee, Dupuis & Trapin (2018); Gerlach & Wang (2020); Wu & Cai (2024); Chen, Koike & Shau (2024).

Forecast evaluation and regimes: Andersen, Bollerslev & Meddahi (2005); Patton (2011); Kupiec (1995); Christoffersen (1998); Diebold & Mariano (1995); Hansen, Lunde & Nason (2011); Fissler & Ziegel (2016); Patton, Ziegel & Chen (2019); Lamoureux & Lastrapes (1990); Mikosch & Stărică (2004); Granger & Hyung (2004); Pesaran & Timmermann (2007).

7. Team verification workflow

This PDF should be used together with the Excel literature evidence matrix. The goal is to prevent summary drift between what a paper actually says and what the final report claims.

1. Open the DOI or publisher link in the spreadsheet.
2. Check the research question, methodology, sample and main findings against the original PDF.
3. Mark the spreadsheet row as Verified, Needs correction, or Read in full.
4. If a claim in this review is stronger than the paper, weaken the review rather than stretching the source.
5. Only after the key rows are verified should the prose literature-review section be treated as submission-ready.

Recommended verification order

Start with: McNeil & Frey (2000); Hansen, Huang & Shek (2012); Engle & Manganelli (2004); Engle & Ng (1993); Žikeš & Baruník (2016); Watanabe (2012); Paul & Sharma (2017/2018); Bee, Dupuis & Trapin (2018); Trapin (2018); Gerlach & Wang (2020); Patton (2011); Fissler & Ziegel (2016); Wu & Cai (2024); Chen, Koike & Shau (2024). These papers most directly determine model design, evaluation, and the defensible research gap.

References and verification links

The list below includes the principal papers discussed in this working review. DOI/publisher links are included for rapid team verification. Journal rankings are intentionally omitted here because current ABDC/ABS/FT classifications should be checked separately in the evidence matrix.

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